Statistics

3rd secondary.

Unit (1)

Correlation and *Regression*

Correlation

Correlation: is a statistical method used to determine the degree and type of relationship between two variables (X, Y).

Types of Correlation:

(1) Direct correlation → is positive (+)

If the two variables increase or decrease together, meaning they move in the same direction.

(2) Inverse correlation → is negative (-)

If the two variable increases while the other decreases, meaning they move in opposite directions.

(3) Degree of Correlation

The degree of correlation is measured by the correlation coefficient (r)

Properties of the Correlation Coefficient (r):

r is (+) Direct correlation: $r \in]0,1]$

r is (-) Inverse correlation: $r \in [-1,0[$

$r \in [-1, 1] \rightarrow -1 \leq r \leq 1$ where the value of the correlation coefficient ranges between -1, 1

(1) Person's correlation coefficient

Suppose that we have two variables X , Y the number of each is (n) then:

$$r=\frac{n\sum xy-(\sum x)(\sum y)}{\sqrt{n\sum x^2-(\sum x)^2}\times\sqrt{n\sum y^2-(\sum y)^2}}$$

Properties of the correlation coefficient

(1) $-1 \le r \le 1$

(2) if $r=0$ then No linear correlation

(3) If $r = 1$ then Direct perfect correlation

(4) If $r = -1$ then inverse perfect correlation

(5) If $0 < r \le 0.4$ then Direct weak correlation

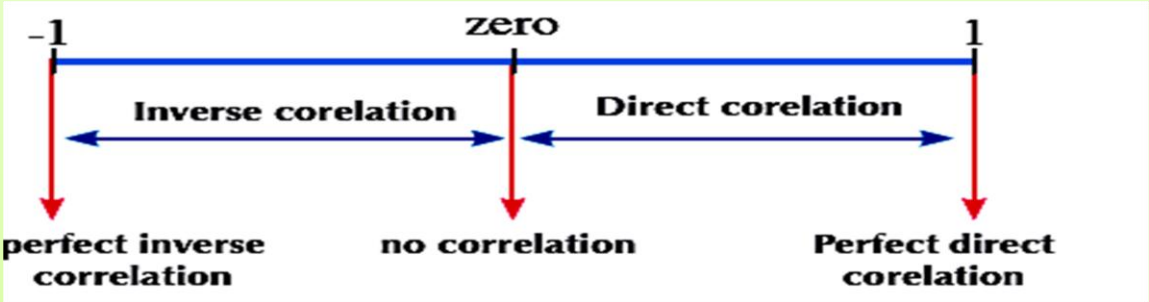
(6) If $0.4 < r \le 0.6$ then Direct moderate correlation.

(7) If $0.6 < r \le 0.99$ then Direct strong correlation

(8) If $-0.4 < r \le 0$ then inverse weak correlation

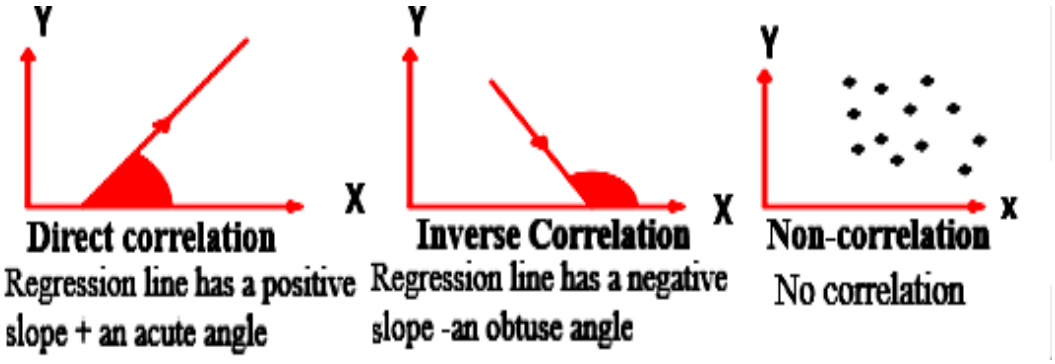
(9) If $-0.6 \le r < -0.4$ then inverse moderate correlation

(10) If $-0.99 \le r < -0.6$ then inverse moderate correlation



Scatter Plot:

A scatter plot is a graphical representation of a set of ordered pairs (X,Y) to describe the relationship between two variables. It helps in identifying the type of correlation between X and Y.



Important Notes:

(1)If all points lie on the regression line then the correlation is perfect

(2)The closer the points are to the regression line, the stronger the correlation (whether direct or inverse), and vice versa

(3) $\bar{X} = \frac{\sum X}{n}$ *then* $\sum x = n\bar{X}$

(4) $\bar{Y} = \frac{\sum Y}{n}$ *then* $\sum y = n\bar{Y}$

Example(1)

Calculate Pearson's correlation coefficient of the following data $\sum x = 68$, $\sum y = 36$, $\sum xy = 348$

$\sum x^2 = 620$, $\sum y^2 = 204$, $n = 8$

Solution

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{n\sum x^2 - (\sum x)^2} \; X \sqrt{n\sum y^2 - (\sum y)^2}}$$

$$r = \frac{8(348) - (68)(36)}{\sqrt{8(620) - (68)^2} \; X \sqrt{8(204) - (36)^2}} = 1$$

< perfect direct >

Example(2) Calculate Pearson's correlation coefficient of the following data:

$\sum x = 5790$, $\sum y = 322$, $\sum xy = 188600$

$\sum x^2 = 3388100$, $\sum y^2 = 10540$, $n = 10$

Solution

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{n\sum x^2 - (\sum x)^2} \; X \sqrt{n\sum y^2 - (\sum y)^2}}$$

$$r = \frac{10(188600) - (5790)(322)}{\sqrt{10(3388100) - (5790)^2} \; X \sqrt{10(10540) - (322)^2}} =$$

$r = 0.87$

<direct strong>

Example(3) If $\Sigma X = 50$, $\Sigma Y = 60$, $\Sigma XY = 260$, $\sum X^2 = 310$, $\sum Y^2 = 500$, $n = 10$, then Pearson's correlation coefficient is:..... (2022)

(a) 0.44 (b) -0.44 (c) -0.34 (d) -0.063

Solution

$$r=\frac{10(260)-(50)(60)}{\sqrt{10(310)-(50)^2}\ X\sqrt{10(500)-(60)^2}} = -0.44$$

Example(4) Calculate Pearson's correlation coefficient of the following data

$\sum x = 92$, $\sum y = 36$, $\sum xy = 372$, $\sum x^2 = 1100$, $\sum y^2 = 204$, $n = 8$

(a) -1 (b) -0.8 (c) 0.8 (d) 1

Solution

$$r=\frac{8(372)-(92)(36)}{\sqrt{8(1100)-(92)^2}\times\sqrt{8(204)-(36)^2}} = -1$$

Example(5) try to solve

Calculate Pearson's correlation coefficient of the following data $\sum x = 220$, $\sum y = 140$, $\sum xy = 2658$ $\sum x^2 = 5486$, $\sum y^2 = 2292$, $n = 10$

معلومه حلوه

(1)If $n\sum xy-(\sum x)(\sum y)>0$ then correlation is direct

(2) If $n\sum xy-(\sum x)(\sum y)<0$ then correlation is inverse

(3) If $n\sum xy-(\sum x)(\sum y)=0$ then no correlation

Example(6) If $n=10$, $\Sigma X=20$, $\Sigma Y=50$, $\Sigma XY=150$, then the correlation between X and Y is:

(a) direct (b) invers (c) perfect (d) non-correlation

Solution

$$n\sum xy-(\sum x)(\sum y)$$
$$=\quad 10\times 150-20\times 50=500>0$$

(a) direct

Example(7) If the correlation between X and Y is nonexistent and $\Sigma X=40$, $\Sigma Y=50$, $\Sigma XY=200$, then $n=$

(a) 20 (b) 10 (c) 50 (d) 15

Solution

$$n\sum xy-(\sum x)(\sum y)=0$$

$$then\quad 200n-50\times 40=0\quad then\ n=10$$

Example(8) in a statistical study to find the relation between two variables x and y, we have got the next data: $\bar{X} = 8$, $\bar{Y} = 10$, $\sum xy = 870$, n =10 , $\sum x^2 = 665$, $\sum y^2 = 1440$ Find linear correlation coefficient

Solution

$\because \bar{X} = 8 \qquad \therefore \sum X = n\bar{X} = 10 \times 8 = 80$

$\because \bar{Y} = 10 \qquad \therefore \sum Y = n\bar{Y} = 10 \times 10 = 100$

$$\therefore r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{n\sum x^2 - (\sum x)^2} \; X \sqrt{n\sum y^2 - (\sum y)^2}}$$

$$r = \frac{10(870) - (80)(100)}{\sqrt{10(665) - (80)^2} \; X \sqrt{10(1440) - (1000)^2}} = 0.7$$

Example(9) if $\bar{X} = 5$, $\bar{Y} = 4$, $\sum xy = 231$, n =10 , $\sum x^2 = 298$, $\sum y^2 = 176$ then the pearson's correlation coefficient between the two variables x and y =

(a) 0.47 (b) -0.47 (c) 0.4 (d) 1

Solution

Example (10)

From the data of the next table

x	1	3	4	6	7	9
y	6	4	4	3	2	1

Calculate Pearson's correlation coefficient between the values of x and y and show its type.

Solution:

X	Y	X^2	y^2	XY
1	6	1	36	6
3	4	9	16	12
4	4	16	16	16
6	3	36	9	18
7	2	49	4	14
9	1	81	1	9
$\sum x=30$	$\sum y=20$	$\sum x^2=192$	$\sum y^2=82$	$\sum xy=75$

$$\therefore r=\frac{n\sum xy-(\sum x)(\sum y)}{\sqrt{n\sum x^2-(\sum x)^2}\ X\sqrt{n\sum y^2-(\sum y)^2}}$$

$$r=\frac{6(75)-(30)(20)}{\sqrt{6(192)-(30)^2}\ X\sqrt{6(82)-(20)^2}}=$$

$$-0.985$$

< strong inverse>

Example (11)

From the data of the next table

x	20	23	24	25	28	30
y	35	31	30	27	29	28

Calculate Pearson's correlation coefficient between the values of x and y and show its type.

Solution:

X	Y	X^2	y^2	XY
20	35			
23	31			
24	30			
25	27			
28	29			
30	28			
$\sum x=$	$\sum y$	$\sum x^2 =$	$\sum y^2 =$	$\sum xy=$

$$\therefore r=\frac{n\sum xy-(\sum x)(\sum y)}{\sqrt{n\sum x^2-(\sum x)^2}\ X\sqrt{n\sum y^2-(\sum y)^2}}$$

r=

CHECK your answer by using the calculator: mode -- 3 – 2 data AC -SHIFT-1-5-3

Example (12)

From the data of the next table

x	3	1	6	4	3	8
y	7	4	5	8	6	7

Calculate Pearson's correlation coefficient between the values of x and y and show its type.

Solution:

X	Y	X^2	y^2	XY
3	7			
1	4			
6	5			
4	8			
3	6			
8	7			
$\sum x=$	$\sum y$	$\sum x^2 =$	$\sum y^2 =$	$\sum xy=$

$$\therefore r=\frac{n\sum xy-(\sum x)(\sum y)}{\sqrt{n\sum x^2-(\sum x)^2}\ X\sqrt{n\sum y^2-(\sum y)^2}}$$

r=

CHECK your answer by using the calculator: mode -- 3 – 2 data AC -SHIFT-1-5-3

Example (13)

From the data of the next table

x	3	1	6	4	3	8
y	7	4	5	8	6	7

Calculate Pearson's correlation coefficient between the values of x and y and show its type.

Solution:

X	Y	X^2	y^2	XY
3	7			
1	4			
6	5			
4	8			
3	6			
8	7			
$\sum x=$	$\sum y$	$\sum x^2 =$	$\sum y^2 =$	$\sum xy=$

$$\therefore r=\frac{n\sum xy-(\sum x)(\sum y)}{\sqrt{n\sum x^2-(\sum x)^2}\ X\sqrt{n\sum y^2-(\sum y)^2}}$$

r=

CHECK your answer by using the calculator: mode -- 3 – 2 data AC -SHIFT-1-5-3

Example (14)

From the data of the next table

x	19	18	7	10	4	12	15	5
y	12	10	7	9	6	13	14	12

Calculate Pearson's correlation coefficient between the values of x and y and show its type.

Solution:

X	Y	X^2	y^2	XY
19	12			
18	10			
7	7			
10	9			
4	6			
12	13			
15	14			
5	12			
$\sum x=$	$\sum y$	$\sum x^2 =$	$\sum y^2 =$	$\sum xy=$

$\therefore r=$

CHECK your answer by using the calculator: mode -- 3 – 2 data AC -SHIFT-1-5-3

Exams

(1) The correlation coefficient of a digital scale whose value ranges between

(a) [0 , 1] (b)]- 1 , 1[(c) [- 1 , 1] (d) [- 1 , 1] – {0}

Solution

Ans:(C)

(2) The strongest inverse correlation coefficient of the following is:

(a) – 0.2 (b) – 0.5 (c) – 0.7 (d) – 0.8

Solution

Ans:(d)

(3) The weakest correlation coefficient of the following is:

(a) - 1.2 (b) - 0.7 (c)0.12 (d) 0.9

Solution

Ans:(C)

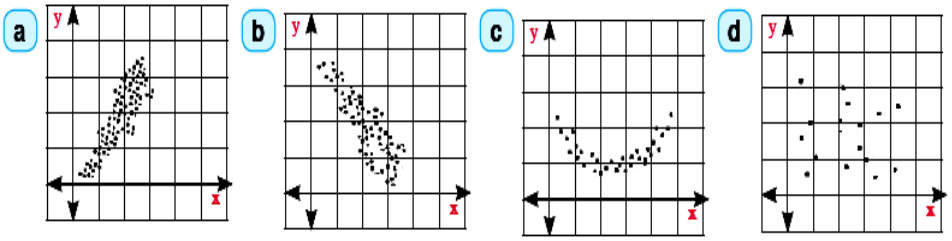
(4) If all the points in a scatter diagram lie on a straight line, the correlation coefficient between the two variables equals:

(a)± 1 (b) 0 (c) 12 (d) 3

Solution

Ans:(a)

(5) The scatter diagram representing a direct correlation is figure



Solution

Ans:(a)

(6) if $\frac{n\sum xy}{(\sum x)(\sum y)} = 1$, the correlation is:

(a) direct (b) inverse (c) perfect (d) non-correlation

Solution

$$\frac{n\sum xy}{(\sum x)(\sum y)} = 1 \ \textit{then} \ n\sum xy = (\sum x)(\sum y)$$

Then $n\sum xy - (\sum x)(\sum y) = 0$

non-correlation

(7) if $\sum x = 68$, $\sum y = 36$, $\sum xy = 348$

$\sum x^2 = 620$, $\sum y^2 = 204$, $n = 8$

then the pearson's correlation coefficient between the two variables x and y =

(a) -2.8 (b) -0.8 (c)0.8 (d)1

Solution

Ans:(d)

H.W(1)

First: Choose the correct answer:

(1) The strongest correlation coefficient of the following is:
(a) – 0.94 (b) zero (c) 0.5 (d) 0.85

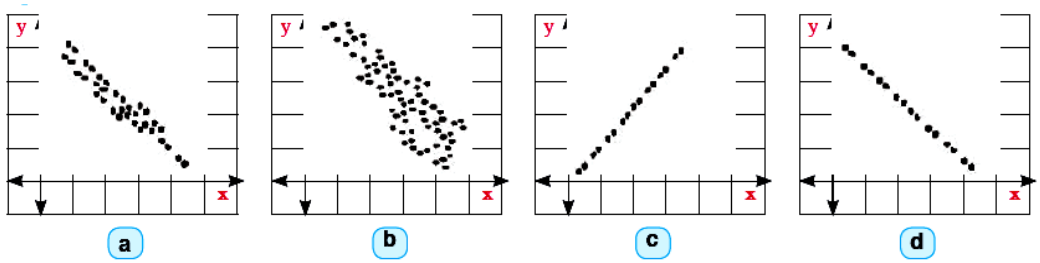
(2) The strongest inverse correlation of the following is:
(a) – 0.2 (b) – 0.5 (c) – 0.7 (d) – 0.8

(3) The weakest correlation coefficient of the following is:
(a) - 1.2 (b) - 0.7 (c) 0.12 (d) 0.9

(4) One of the next numbers can represent the strongest inverse correlation coefficient between two variables:
(a) 0.3 (b) 0.9 (c) - 1.1 (d) - 0.95

(5) The correlation coefficient of a digital scale whose value ranges between
(a) [0 , 1] (b)]- 1 , 1[(c) [- 1 , 1] (d) [- 1 , 1] – {0}

(6) the figure shows a strong inverse correlation between x and y is figure:



(7) The relation between the square side length and its area is a .....correlation:
(a) strong direct (b) strong inverse
(c) perfect direct (d) perfect inverse

(8) If the two variables increase or decrease together, the correlation between them is
(a) direct. (b) inverse. (c) non-linear. (d) nihilistic

(9) if $\sum x = 68$, $\sum y = 36$, $\sum xy = 348$
$\sum x^2 = 620$, $\sum y^2 = 204$, n = 8
then the pearson's correlation coefficient between the two variables x and y =
(a) -2.8 (b) -0.8 (c)0.8 (d)1

(10) If $\Sigma X = 50$, $\Sigma Y = 60$, $\Sigma XY = 260$,
$\sum X^2 = 310$, $\sum Y^2 = 500$, n = 10, then Pearson's correlation coefficient is:..... (2022)
(a) 0.44 (b) -0.44 (c) -0.34 (d) -0.063

(11) Calculate Pearson's correlation coefficient of the following data $\sum x = 92$, $\sum y = 36$, $\sum xy = 372$,
$\sum x^2 = 1100$, $\sum y^2 = 204$, n = 8
(a) -1 (b) -0.8 (c) 0.8 (d) 1

(12) if $\bar{X} = 5$, $\bar{Y} = 4$, $\sum xy = 231$, n =10 , $\sum x^2 = 298$, $\sum y^2 = 176$ then the pearson's correlation coefficient between the two variables x and y =
(a) 0.47 (b) -0.47 (c)0.4 (d)1

(13) If the correlation between X and Y is nonexistent and $\Sigma X = 40$, $\Sigma Y = 50$, $\Sigma XY = 200$, then n =
(a) 20 (b) 10 (c) 50 (d) 15

(14) The correlation coefficient between the side length of a square and its perimeter is:
(a) 0.25 (b) 1 (c) 0.8 (d) – 1

(2) Calculate Pearson's correlation coefficient of the following data $\sum x = 92$, $\sum y = 36$, $\sum x\,y = 372$ $\sum x^2 = 1100$, $\sum y^2 = 204$, $n = 8$

(3) Calculate Pearson's correlation coefficient of the following data $\sum x = 220$, $\sum y = 140$, $\sum x\,y = 2658$ $\sum x^2 = 5486$, $\sum y^2 = 2292$, $n = 10$

(4) Calculate Pearson's correlation coefficient of the following data: $\sum x = 5790$, $\sum y = 322$, $\sum x\,y = 188600$ $\sum x^2 = 3388100$, $\sum y^2 = 10540$, $n = 10$

(5) in a statistical study to find the relation between two variables x and y, we have got the next data: $\bar{X} = 8$, $\bar{Y} = 10$, $\sum x\,y = 870$, $n = 10$, $\sum x^2 = 665$, $\sum y^2 = 1440$ Find linear correlation coefficient

(6) if $\bar{X} = 5$, $\bar{Y} = 4$, $\sum x\,y = 231$, $n = 10$, $\sum x^2 = 298$, $\sum y^2 = 176$ FIND the pearson's correlation coefficient between the two variables x and y =

(7) Calculate Pearson's correlation coefficient of the following data: $\sum x = 49$, $\sum y = 45$, $\sum x\,y = 320$ $\sum x^2 = 359$, $\sum y^2 = 303$, $n = 7$

(8) From the data of the next table

x	12	10	14	11	12	9
y	18	17	23	19	20	15

Calculate Pearson's correlation coefficient between the values of x and y and show its type.

(9) From the data of the next table

x	3	1	6	4	3	8
y	7	4	5	8	6	7

Calculate Pearson's correlation coefficient between the values of x and y and show its type.

(10) From the data of the next table

x	7	7	8	3	7	11
y	8	4	12	2	10	1

Calculate Pearson's correlation coefficient between the values of x and y and show its type.

(11) From the data of the next table

x	6	5	7	8	10	6	7
y	4	7	5	6	8	7	8

Calculate Pearson's correlation coefficient between the values of x and y and show its type

(12) From the data of the next table

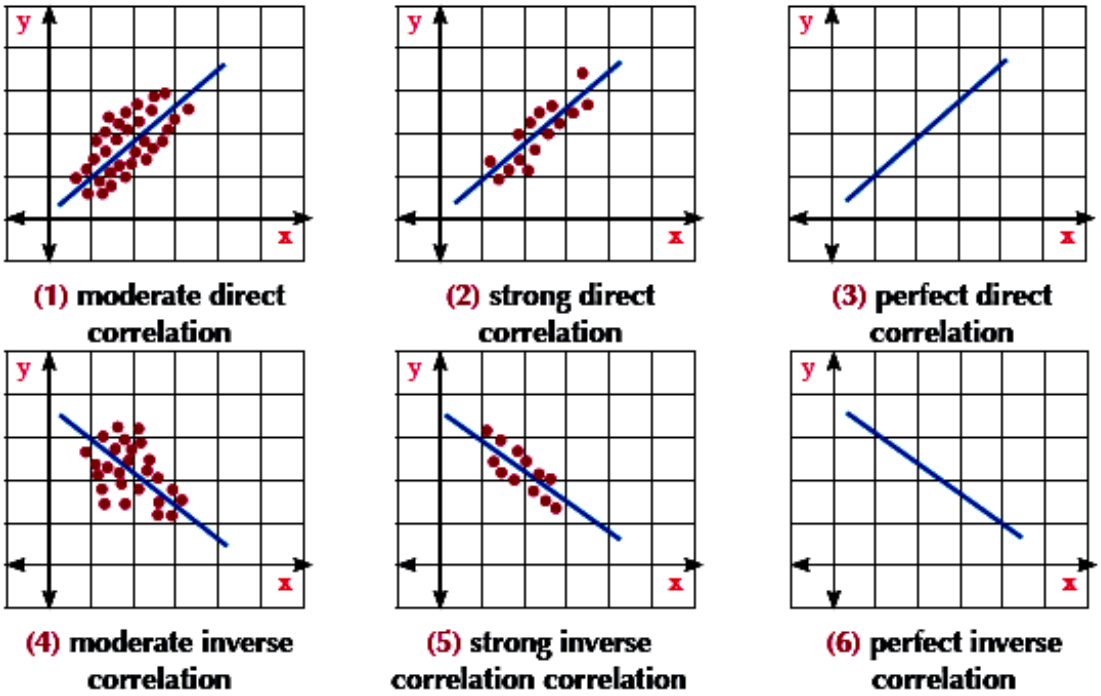
x	19	18	7	10	4	12	15	5
y	12	10	7	9	6	13	14	12

Calculate Pearson's correlation coefficient between the values of x and y and show its type.

(2)Regression

Definition

Regression is a statistical method by which the value of a variable can be estimated in terms of the other variable.



❖ **Regression line equation: y = a+ b x** where :

a is the length of y-intercept

b is the regression coefficient of y on x and it expresses the slope of the regression line on the positive direction of X-axis.

$$b=\frac{n\sum xy-(\sum x)(\sum y)}{n\sum x^2-(\sum x)^2} \quad ,a=\frac{\sum y-b\sum x}{n}$$

❖ **The regression line equation is used for:**

(1) Predicting the value of y if the value of x is known

(2) Identifying the error which can be determined by the relation:

Error = | Table value - the value satisfying the regression equation

Example(1)

in a study to find the relation between the income (x) and consumption (y) in thousand pounds, the results show that

:$\sum x = 100$, $\sum y = 120$, $\sum xy = 516$

$\sum x^2 = 410$, $\sum y^2 = 720$, n =40

(1) find the regression line equation.

(2) predict the value of consumption (y) when the income is LE .700

Solution

(1)

$$b=\frac{n\sum xy-(\sum x)(\sum y)}{n\sum x^2-(\sum x)^2}=\frac{40\times516-100\times120}{40\times410-(100)^2}=1.35$$

$$a=\frac{\sum y-a\sum x}{n}=\frac{120-1.35\times100}{40}=-0.375$$

The regression line equation of y on x is

y = a+bx =1.35x-0.375

(2) at x=700

∴ y=1.35×700-0.375=907.5

Example(2)

$\sum x = 33$, $\sum y = 24$, $\sum x\,y = 135$
$\sum x^2 = 196$, $\sum y^2 = 106$, n = 6 find the equation of the regression line y on x ,then estimate the value of (y) at x=13

Solution

(1) $a=\frac{n\sum xy-(\sum x)(\sum y)}{n\sum x^2-(\sum x)^2}=\frac{6\times135-33\times24}{6\times196-(33)^2}=0.207$

$b=\frac{\sum y-a\sum x}{n}=\frac{24-0.207\times33}{6}=2.86$

The regression line equation of y on x is

$y = ax + b = 0.207x+2.86$

(2) at x=13

$\therefore y=0.207\times13+2.86= 6.37$

The regression line equation of y on x is

$y = bx + a = 0.207x+2.86$

(2) at x=13 $\therefore y=0.207\times13+2.86= 6.37$

Example(3)Try to solve

$\sum x = 30$, $\sum y = 40$, $\sum x\,y = 162$
$\sum x^2 = 210$, $\sum y^2 = 304$, n = 6
find regression line equation

Example(4) in a statistical study to find the relation between two variables x and y, we have got the next data:
$\bar{X} = 8$, $\bar{Y} = 10$, $\sum xy = 870$, n =10, $\sum x^2 = 665$, $\sum y^2 = 1440$
(1) linear correlation coefficient
(2) regression line equation of y on x

Solution

(1) $\because \bar{X} = 8$ $\therefore \sum X = n\bar{X} = 10 \times 8 = 80$

$\because \bar{Y} = 10$ $\therefore \sum Y = n\bar{Y} = 10 \times 10 = 100$

$$\therefore r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{n\sum x^2 - (\sum x)^2} \; X \sqrt{n\sum y^2 - (\sum y)^2}}$$

$$r = \frac{10(870) - (80)(100)}{\sqrt{10(665) - (80)^2} \; X \sqrt{10(1440) - (1000)^2}} = 0.7$$

(2) $$b = \frac{n\sum xy - (\sum x)(\sum y)}{n\sum x^2 - (\sum x)^2} = \frac{10(870) - (80)(1000}{10(665) - (80)^2} = 2.8$$

$$a = \frac{\sum y - b\sum x}{n} = \frac{100 - 2.8 \times 80}{10} = -12.4$$

The regression line equation of y on x is

$y = bx + a = 2.8x - 12.4$

Example(5) if $\sum x = 10$, $\sum y = 32$ and n=4 , if the regression line equation is: y = a +2x then a=....

(a) 1 (b) 0 (c) 3 (d) 2

Solution

$$a=\frac{\sum y-b\sum x}{n}=\frac{32-2\times10}{4}=3$$ Ans:(C)

Example(6)

From the data of the following table (2025)

X	5	8	10	14	16	20
y	4	6	9	11	12	15

Then the equation of regression line y on x is y=.....

(a) 0.703 - 0.723 x (b) 0.723 - 0.703 x

(c) 0.703 + 0.723 x (d) 0.723 + 0.703 x

Solution

X	Y	X^2	XY
5	4	25	20
8	6	64	48
10	9	100	90
14	11	196	154
16	12	256	192
20	15	400	300
$\sum x = 73$	$\sum y = 57$	$\sum x^2 = 1041$	$\sum xy = 804$

(1) $$b=\frac{n\sum xy-(\sum x)(\sum y)}{n\sum x^2-(\sum x)^2}=\frac{6\times804-73\times57}{6\times1041-(73)^2}=0.723$$

$$a=\frac{\sum y-a\sum x}{n}=\frac{57-0.723\times73}{6}=0.703$$

The regression line equation of y on x is

y = bx + a= 0.703 + 0.723 x

Example (7)

From the data of the next table

x	1	3	4	6	7	9
y	6	4	4	3	2	1

First: find the regression line equation.

Solution

X	Y	X^2	y^2	XY
1	6	1	36	6
3	4	9	16	12
4	4	16	16	16
6	3	36	9	18
7	2	49	4	14
9	1	81	1	9
$\sum x=30$	$\sum y=20$	$\sum x^2=192$	$\sum y^2=82$	$\sum xy=75$

$$b=\frac{n\sum xy-(\sum x)(\sum y)}{n\sum x^2-(\sum x)^2}=\frac{6(75)-(30)(20)}{6(192)-(30)^2}=-0.595$$

$$a=\frac{\sum y-b\sum x}{n}=\frac{20-(-0.595)\times 30}{6}=6.31$$

The regression line equation of y on x is

$$y=bx+a==-0.595\,x-6.31$$

Example (8)

From the data of the next table

x	5	8	10	14	16	20
y	4	6	9	11	12	15

find the regression line equation and predict the value of y when x = 12

Solution

X	Y	X^2	y^2	XY
5	4			
8	6			
10	9			
14	11			
16	12			
20	15			
$\sum x=$				

$$b=\frac{n\sum xy-(\sum x)(\sum y)}{n\sum x^2-(\sum x)^2}=$$

$$a=\frac{\sum y-b\sum x}{n}=$$

The regression line equation of y on x is

$$y=bx+a=$$

Example (9)

From the data of the next table

x	20	33	30	40	13	15	26	25
y	7	8	9	11	4	5	8	9

(a) predict the value of y when x = 35

(b) find the error in y if x= 30

Solution

X	Y	X^2	y^2	XY
20	7			
33	8			
30	9			
40	11			
13	4			
15	5			
26	8			
27	9			

$$b=\frac{n\sum xy-(\sum x)(\sum y)}{n\sum x^2-(\sum x)^2}= \qquad a=\frac{\sum y-b\sum x}{n}=$$

The regression line equation of y on x is

y = bx + a=

(2)To find the error in the production if you know that x= 30 s:

y =

Error = |Table value - the value satisfying the regression equation|

Exams

(1)if the regression line equation is: y = 2 + 0.5x then the value of y expected when x = 6 is:
(a) 4 (b) 5 (c) 7 (d) 8

Solution

y=2+0.5(6)=5 Ans:(b)

(2)If the regression line equation of y on x is
y = 0.2 x + 3 and the value table of y when x = 5 is 4.6 , then the error in the value of y = ...
(a) 4 (b) 0.6 (c) -0.6 (d) 0.4

Solution

Ans:(b)Y=0.2(5)+3=4
Error = | Table value - the value satisfying the regression equation | = | 4.6-4|=0.6

(3)If the regression equation is $\hat{Y}$ =0.4x+2 and the actual table value of y when x = 3 is 4, the error in y is:
(a) 0.2 (b) 0.6 (c) 0.8
(d) 3.5

Solution

At x=3 then Y=0.4(3)+3=4 .2
Error = | Table value - the value satisfying the regression equation | = | 4-4.2|=0.2

(4)If the regression line equation y on x is :
y = 3-x, then the correlation between the values of x and the values of y is:
(a) nihilistic (b) direct (c) inverse (d) non correlation

Solution

Ans:(c)

(5)If the two points (6,5) and (3,8) lie on the regression line y on x, then the type of the correlation between x and y is ....
(a) direct (b) inverse (c) perfect (d) nihilistic

Solution

slope=$\frac{8-5}{3-6} = -1$ Ans:(b)

(6)if $\sum x$ =10 , $\sum y$ = 32 and n=4 , if the regression line equation is: y = a +2x then a=....
(a) 1 (b) 0 (c) 3 (d) 2

Solution

a=$\frac{\sum y - b\sum x}{n} = \frac{32-2\times 10}{4} = 3$

(7)From the data of the following table

X	6	5	7	8	10	6	7
y	4	7	5	6	8	7	8

if the regression line equation is: y = 4.2 + 0.3x then the error in the value of y when x = 10 is:
(a) -2.8 (b) -0.8 (c)0.8 (d)1. 8

Solution

Y=4.2+0.3(10)=7.2
Error = | Table value - the value satisfying the regression equation |= | 8-7.2|=0.8

(8)If the slope of the regression equation Y on X is 0.25, and n=4 , ΣX=12, ΣY=0, then the y-intercept is:
(a) 2 (b) 1.75 (c) -0.75 (d) 3

Solution

a=$\frac{\sum y - b\sum x}{n} = \frac{0-0.25\times 12}{4} = -0.75$
the regression equation is Y =0.25x-0.75
the y-intercept is: -0.75

H.W(2)

(1) the statistical equation of the regression line equation where b is the regression coefficient is:
(a) $y = a x + b$ (b) $y = a + b x$
(c) $x = a y + b$ (d) $x = a + b y$.

(2) if the regression line equation is: $y = 3 + 0.5x$ then the value of y expected when $x = 4$ is:
(a)4 (b) 5 (c) 7 (d) 8

(3)if the regression line equation is: $y = 2 + 0.5x$ then the value of y expected when $x = 10$ is:
(a) 4 (b) 5 (c) 7 (d) 8

(4) if the two points (11.5 , 10) and (6.5 , 5) lie on the regression line y on x, then the correlation between x and y is:
(a) direct (b) inverse (c) perfect (d) nihilistic

(5) if the two points (5 , 13) ,and (14 , 4) lie on the regression line y on x, then all the following points lie on the same line except:
(a) (15 , 5) (b) (10 , 8) (c) (6 , 12) (d) (5 , 13)

(6)If the regression line equation of y on x is
$y = 0.2 x + 3$ and the value table of y when $x = 5$ is 4.6 , then the error in the value of y = ...
(a) 4 (b) 0.6 (c) -0.6 (d) 0.4

(7) if all the points in a scatter diagram lie on a straight line whose slope is negative, then the correlation coefficient between x and y equals:
(a) 1 (b) 0 (c) -0.5 (d) -1

(8) if all the points in a scatter diagram lie on a straight line whose slope is positive, then the correlation coefficient between the two variables equals:
(a) -1 (b) 0 (c) 0.5 (d) 1

(9) the variable required to estimated in the regression line equation is called .....variable

(a) independent (b) dependent (c) direct (d)inverse

(10) if all the points in a scatter diagram lie on a straight line ,then the correlation between the two variable may be equal

(a) $-\frac{1}{2}$ (b) 0 (c) -1 (d) $\frac{3}{4}$

(11)if $\sum x$ =10 , $\sum y$ = 32 and n=4 , if the regression line equation is: y = a +2x then a=....

(a) 1 (b) 0 (c) 3 (d) 2

(12)If the regression line equation y on x is :
y = 5-x, then the correlation between the values of x and the values of y is:

(a) nihilistic (b) direct (c) inverse (d) non correlation

(13)if $\sum x$ =79 , $\sum y$ = 179 $\sum$ x y = 2899,$\sum x^2$ = 1331 , , n = 7 ,then regression line equation of y on x is.....

(a) y=2x+3 (b) y=2x-3 (c) y=3x+2 (d) y=3x-2

(14)From the data of the following table

X	6	5	7	8	10	6	7
y	4	7	5	6	8	7	8

if the regression line equation is: y = 4.2 + 0.3x then the error in the value of y when x = 10 is:

(a) -2.8 (b) -0.8 (c)0.8 (d)1. 8

(15)From the data of the following table

X	6	5	7	8	10
y	4	7	5	6	8

if the error at x=8 is 0.3 then the value satisfying the regression equation may be equals in the value of y when x = 10 is:

(a) 6 (b) 6.6 (c)6.3 (d)10

Second: Answer the following questions:

(1)From the data of the next table

x	5	8	10	14	16	20
y	4	6	9	11	12	15

find the regression line equation and predict the value of y when x = 12

(2)From the data of the next table

x	1	3	4	6	7	9
y	6	4	4	3	2	1

find the regression line equation.

(3)From the data of the next table

x	20	33	30	40	13	15	26	25
y	7	8	9	11	4	5	8	9

a) predict the value of y when x = 35

b) find the error in y if x= 30

(4)$\sum x = 30$, $\sum y = 40$, $\sum x y = 162$

$\sum x^2 = 210$, $\sum y^2 = 304$, n = 6 find regression line equation

(5)in a study to find the relation between the income (x) and consumption (y) in thousand pounds, the results show that

:$\sum x = 100$, $\sum y = 120$, $\sum x y = 516$ $\sum x^2 = 410$, $\sum y^2 = 720$, n =40

(1) find the regression line equation.

(2) predict the value of consumption (y) when the income is LE .900

(6)$\sum x = 33$, $\sum y = 24$, $\sum x y = 135$

$\sum x^2 = 196$, $\sum y^2 = 106$, n = 6 find the equation of the regression line y on x ,then estimate the value of (y) at x=13

(7)in a statistical study to find the relation between two variables x and y, we have got the next data:

$\bar{X} = 8$,$\bar{Y} = 10$, $\sum x y = 870$, n =10,$\sum x^2 = 665$, $\sum y^2 = 1440$

1)linear correlation coefficient

2) regression line equation of y on x

(3) Spearman's rank correlation coefficient.

$$r=1-\frac{6\sum D^2}{n(n^2-1)}=1-\frac{6\sum D^2}{(n^3-n)}$$

Example (1) if D is the difference between the ranks of each corresponding values of x ,y and $\sum D^2 = zero$ then the correlation coefficient between x ,y=

(a) 1 (b) 0 (c) -0.6 (d) -1

Solution

$$r=1-\frac{6\sum D^2}{n(n^2-1)}=1-0=1$$ Ans:(a)

Example (2) if Dis the difference between the ranks of each corresponding values of x ,y and $\sum D^2 = 35$ and n=6 then the correlation coefficient between x ,y=

(a) 1 (b) 0,5 (c) -0.5 (d) 0

Solution

$$r=1-\frac{6\sum D^2}{n(n^2-1)}=1-0=1$$

Ans:(d) (2023)

Example (3) If $\Sigma D^2= 60$ and $n^3 - n - 504=0$, then Spearman's rank correlation coefficient r =.....

(a) 0.2 (b) 0.3 (c) 0.5 (d) 0.98

Solution

$$r=1-\frac{6\sum D^2}{n(n^2-1)}=1-\frac{6\times60}{504}=0.3$$

Ans:(b)

Example (4)

Calculate Spearman's rank correlation coefficient X , Y

X	15	25	35	35	35	25	15
Y	6	6	9	8	7	8	5

Solution

:

Order of X:- *15 15 25 25 35 35 35*

1 2 3 4 5 6 7

$\frac{1+2}{2}=1.5$ $\frac{3+4}{2}=3.5$ $\frac{5+6+7}{3}=6$

Rank of X: *1.5 1.5 3.5 3.5 6 6 6*

Order of Y:- *5 6 6 7 8 8 9*

1 2 3 4 5 6 7

Rank of Y: *1 2.5 2.5 4 5.5 5.5 7*

Ranks of X	Ranks of Y	D	D^2
1.5	2.5	1	1
3.5	2.5	1	1
6	7	1	1
6	5.5	0.5	0.25
6	4	2	4
3.5	5.5	2	4
1.5	1	0.5	0.25

$$r=1-\frac{6\sum D^2}{n(n^2-1)}=1-\frac{6\times 11.5}{7(7^2-1)}=0.797$$

(Direct strong)

Example (5) The following table gives the scores of 8 students in history and geography. Calculate the correlation coefficient

History	weak	Pass	pass	weak	v.g	Exc.	good	pass
geography	pass	good	good	pass	good	v.g	Exc.	weak

Solution

Order of X:-	*W*	*W*	*P*	*P*	*P*	*G*	*V.G*	*EXC*
	1	*2*	*3*	*4*	*5*	*6*	*7*	*8*
Rank of X:	*1.5*	*1.5*	*4*	*4*	*4*	*6*	*7*	*8*

$$\frac{1+2}{2}=1.5 \qquad \frac{3+4+5}{3}=4$$

Order of Y:-	*W*	*P*	*P*	*G*	*G*	*G*	*V.G*	*EXC.*
	1	*2*	*3*	*4*	*5*	*6*	*7*	*8*
Rank of Y:	*1*	*2.5*	*2.5*	*5*	*5*	*5*	*7*	*8*

Ranks of X	Ranks of Y	D	D^2
1.5	2.5	-1	1
4	5	-1	1
4	5	-1	1
1.5	2.5	-1	1
7	5	2	4
8	7	1	1
6	8	-2	4
4	1	3	9
			22

$$r=1-\frac{6\sum D^2}{n(n^2-1)}=1-\frac{6\times 22}{8(8^2-1)}=0.74$$

(Direct strong)

Example (6) The following table . Calculate the correlation coefficient

X	Exc.	Pass	good	weak	v.g
Y	weak	V,good	good	Exc.	pass

(a) -1 (b) -0.2 (c)0 (d)1

Solution

Ans:(A)

x	y	Ranks of X	Ranks of Y	D	D^2
EXC	WEAK	5	1	4	16
P	V.G	2	4	-2	4
G	G	3	3	0	0
WEAK	EXC	1	5	-4	16
V.G	P	4	2	2	4
SUM				0	40

$$r=1-\frac{6\sum D^2}{n(n^2-1)} = 1-\frac{6\times 40}{5(25-1)}=-1$$

Example (7)

Calculate Spearman's rank correlation coefficient X , Y

X	10	7	8	7	6	4
Y	5	8	7	9	9	10

Solution

Order of X:- *10 7 8 7 6 4*
1 2 3 4 5 6

Rank of X:

Order of Y:- *5 8 7 9 9 10*
1 2 3 4 5 6

Rank of Y:

Ranks of X	Ranks of Y	D	D^2

$$r=1-\frac{6\sum D^2}{n(n^2-1)}=$$

Example (8) The following table gives the scores of 8 students in history and geography. Calculate the correlation coefficient

History	pass	V .g	EXc	V .g	PASS	pass
geography	good	good	v.good	EX	good	WEAK

Solution

Order of X:-

Rank of X:

Order of Y:-

Rank of Y

Ranks of X	Ranks of Y	D	D^2

$$r = 1 - \frac{6\sum D^2}{n(n^2-1)} =:$$

H.W(3)

(1)Spearman's rank correlation coefficient can be calculated using ....
(B) Descriptive data (A) Numerical data
(C) Both numerical and descriptive data (D) None of the above

(2)if Dis the difference between the ranks of each corresponding values of x ,y and $\sum D^2 = 40$ and n=5 then the correlation coefficient between x ,y=
(a) 1 (b) 0 (c) -0.6 (d) -1

(3) if Dis the difference between the ranks of each corresponding values of x ,y and $\sum D^2 = zero$ then the correlation coefficient® between x ,y=
(a) 1 (b) 0 (c) -0.6 (d) -1

(4) if Dis the difference between the ranks of each corresponding values of x ,y and $\sum D^2 = 35$ and n=6 then the correlation coefficient between x ,y=
(a) 1 (b) 0,5 (c) -0.5 (d) 0

(5) If $\Sigma D^2 = 60$ and $n^3 - n - 504 = 0$, then Spearman's rank correlation coefficient r =.....
(a) 0.2 (b) 0.3 (c) 0.5 (d) 0.98

(6)The following table . Calculate the correlation coefficient

X	Exc.	Pass	good	weak	v.g
Y	weak	V,good	good	Exc.	pass

(a) -1 (b) -0.2 (c)0 (d)1

(7)The following table . Calculate the correlation coefficient

X	10	12	14	16
Y	18	20	21	24

(a) -1 (b) 0.5 (c)0 (d)1

Second: Answer the following questions:

(1) Calculate Spearman's rank correlation coefficient X , Y

X	10	7	8	7	6	4
Y	5	8	7	9	9	10

(2) Calculate Spearman's rank correlation coefficient X , Y

X	7	7	8	3	7	11
Y	8	4	12	2	10	11

(3) Calculate Spearman's rank correlation coefficient X , Y

X	12	10	14	11	12	9
Y	18	17	23	19	20	15

(4) Calculate Spearman's rank correlation coefficient X , Y

X	3	1	6	4	3	8
Y	7	4	5	8	6	7

(5) The following table gives the scores of 8 students in history and geography. Calculate the correlation coefficient

History	pass	V .g	EXc	V .g	PASS	pass
geography	good	good	v.good	EX	good	WEAK

(6) Calculate Spearman's rank correlation coefficient between x and y and show its type.

X	V .g	V .g	good	weak	pass	V .g
Y	good	pass	good	Ex	V .g	pass

(7) Calculate Spearman's rank correlation coefficient between x and y and show its type.

X	EXc	good	v.good	pass	weak	good
Y	good	weak	pass	Exc	V .g	pass

Unit (2)

Advanced Measurements in Statistics

(1)Stem-and-Leaf Data Representation

Remember that

(1)Arithmetic mean $= \frac{\text{sum OF values}}{\text{their number}}$

Example: Find the arithmetic mean of the values 5,11,2,8, 4.

mean $= \frac{5+1+2+8+4.}{5} = 4$

(2) Median : It is the value that is in the middle of a group of visual values in ascending or descending order

For example: 20 ,30 , 17 , 23 ,42

Arrange:17,20,23,30,42 then median =23

For example: 21 , 13 , 24 , 23 , 13 , 27

Arrange:13,13,21,23,24,27 then median $= \frac{21+23}{2} = 22$

(3) Mode : It is the most frequent or common value

Example: Find the mode of the values 8, 4, 3, 7, 4.

Mode =4

(4) The range: it is the difference between the highest and lowest values in a data

Example: Find the range of the values 4, 17, 13, 9, 8, 3, 2, 6.

Range = $Maximum\ Value - Minimum\ Value = 17 - 2 = 15$

Example(1)

The given table shows the marks of 15 students in statistics exam, the maximum mark is 50

36	27	38	50	42	46	42	42
39	40	32	41	30	35	23	

(1) Represent the data by using stem and leaf.
(2) How many students have got more than 40 marks ?
(3) Find the range the mode , and the median for these values.

solution

Arrange: 23 , 27, 30 , 32 , 35 , 36 ,38 ,39,40,41,42,42,42,,46,50

stem	leaves
2	3 7
3	0 2 5 6 8 9
4	0 1 2 2 2 6
5	0

Key 2/7 = 27

(2)Number of the students have got more than 40 marks =6

(3)The range = The greatest ark - the smallest mark
= 50 - 23 = 27

Example(2)

The following data shows the marks of some students in a mathematics exam.

92	78	73	89	86	85	76	81	98	75
73	88	83	83	71	86	82	94	100	83

(a) Represent this data using a stem-and-leaf plot (method).

(b) Calculate the median of these marks

solution

Arrange: 71 , 73, 73 , 75 , 76 , 78 ,81 ,82, 83, 83, 83, 85, 86, 86, 88 ,89 ,92, 94, 98 ,100

stem	leaves
7	1 3 3 5 6 8
8	1 2 3 3 3 5 6 6 8 9
9	2 4 8
10	0

median$=\frac{38+38}{2}=38$ Key 8/2 = 82

Example(3)

Choose the correct answer from the given answers

(1) In the opposite representation: the largest number is .....

(a) 2.71 (b) 23.5 (c) 27.5 (d) 275

stem	leaves
23	4 5
24	4 7 9
25	0 4 8 8
26	3 8 9
27	1 2 5

Key 24.7 = 24|7

(2) The median of the previous representation .....

(a) 25.4 (b) 25.8 (c) 254 (d) 258

(3) The range=27.5-23.4=4.1

Example(4)

The following data represents time of the cycling race in one of the Olympic Games, measured in seconds:

89.4	90.4	87.5	84.3	89.7,	90.3	91.4
91	86.7	84.1	89.2	89.2	86	89.1
88.2	89.5	90.5	90.2	89.2	91.1	88.9

(a) Representation of the data using a stem-and-leaf plot

(b) How long did it take the last competitor to reach the end of the race?

Solution

Arrange:84.1 ,84.3 ,86 ,86.7 ,87.5 ,88.2, 88.9 ,89.1 ,89.2 ,89.2 ,89.4 ,89.5 ,89.7 ,90.2 ,90.3 ,90.4 ,90.5 91 ,91.1 ,91.4 key =87/5=87.5

stem	leaves
84	1 3
86	0 7
87	5
88	2 9
89	1 2 2 4 5 7
90	2 3 4 5
91	0 1 4

(2)The last competitor took 91.4 seconds

Example(5) The following data represents the records of some swimmers have achieved in a 50 meter freestyle swimming race measured in seconds.

25.7 ,27.3, 25.6 , 30 ,31.4 , 28.2 ,29.1
28.6 ,26.4 , 27.3 , 30.6 , 27.4 , 25.9 , 27.6
32, 31.9 , 26.9, 29.1 , 27.3 , 28.4

(1) Represent the data by using stem and leaf

(2) What is the time of the first contestant and the last contestant do need to finish the race

Solution try to solve

Double representation with stem and leaves

Example(6)

The following data represents the maximum and minimum temperatures for Alexandria city over a period of two weeks.

Maximum temperature	19	28	22	29	25	29	32	35	36	34	37	39	41	42
Minimum temperature	13	22	19	18	16	20	21	22	23	23	21	30	32	31

Represent the temperatures by stem and leaves with description of this date and which of these data is more dispersion

Solution

Leaves(max)	stem	Leaves(min)
9	1	3 6 8 9
9 9 8 5 2	2	0 1 1 2 2 3 3
9 7 6 5 4 2	3	0 1 4
2 1	4	

32 = 3|2 Key 13 = 1|3

Example(7)

In a study of the time (in minutes) taken to complete a specific task by two groups of workers, their data were as follows:

Group A	32	35	24	36	44	38	40
Group B	43	33	47	37	39	45	41

(1)Draw a double stem-and-leaves plot for this data

(2) Determine which group is faster in completing the task based on the median and range

Solution try to solve

Example(8)

The following data represents the marks Ahmed and Mahmoud get in a weekly mathematics test. Where the test is out of 50.

Ahmed	37 45 31 36 35 40 41 42 40 45 50 39
Mahmoud	42 23 34 37 41 34 37 45 46 48 45 43

Represents the marks Ahmed and Mahmoud get by using the double stem and leaf. Describe these marks and state which of them more divergent.

Solution try to solve

Exams

(1) By using the opposite stem-and - leaves diagram the median =…

stem	leaves
23	4 5
24	4 7 9
25	0 4 8 8
26	3 8 9
27	1 2 5

key 24 | 7 represents 24.7

(a) 25.4 (b) 25.8 (c) 254 (d) 258

Solution

(2) From the stem and leaves diagram opposite then

the median is .....

a) 16

b) 17

C) 18

d) 20

stem	leaves
0	9
1	0 2 2 2 3 4 5 6 6
2	0 1 1 5 7 8 9
3	1 2 3
Key	1|4 = 14

Solution

(3) In the opposite diagram Which of the following statement is in correct?

a) The range of the set $(A) = 30$

b) The median for the set $(B) = 62$

c) the mode of the set $(A) = 43$

d) set A is more variant than set B

Set B	Leaves	Set A
0	4	1 2 3 3
6 3	5	4 5
2 1	6	2 5
7 6 5 2	7	1

Key 3|5|4 means 54 for set a and 53 for set B

Solution

(4)

7. The data given in the stem and leaves diagram are ...

a) 162,1745,1867

b) 162,174,175,186,187

c) 16.2,17.4,17.5,18.6,18.7

d) 1.62,1.74,1.75,1.86,1.87

Stem	Leaves
16	2
17	4 5
18	6 7

Key: 17|4 = 17.4

Solution

(5)

The given stem and leaf diagram represents the maximum and minimum temperature in ElSharkeya governorate during 5 days, then the difference between the arithmetic mean of the maximum and the arithmetic me the minimum equals .....

(a) 2.6 (b) 8 (c) 27.8 (d)19

Maximum	Stem	Minimum
8	1	5 6
3 2	2	1
4	3	3
2	4	1

7: 8|1|5 means maximum 18 and minimum 15

H.W(1)

(1) Use the stem and leaves to find

(1) the median=

(a) 16 (b) 17 (c) 18 (d) 20

stem	leaves
0	9
1	0 2 2 2 3 4 5 6 6
2	0 1 1 5 7 8 9
3	1 2 3

Key 14 = 1|4

(2) the range .....

(a) 21 (b) 22 (c) 24 (d) 25

(3) The mode=27.5-23.4=

(a) 12 (b) 16 (c) 21 (d) 33

(2) Choose the correct answer from the given answers

(1) In the opposite representation: the largest number is .....

(a) 2.71 (b) 23.5 (c) 27.5 (d) 275

stem	leaves
23	4 5
24	4 7 9
25	0 4 8 8
26	3 8 9
27	1 2 5

Key 24.7 = 24|7

(2) The median of the previous representation .....

(a) 25.4 (b) 25.8 (c) 254 (d) 258

(3) The range=

(a) 1.4 (b) 4.1 (c) 4.2 (d) 4

(3)

first: the smallest number is .....

(a) 65.1 (b) 6.51 (c) 0.6516 (d) 651

stem	leaves
65	1 3
66	4 5 6
67	7 8
68	0 2 7
69	3 6

Key 66/4 = 66.4

Second: The median is .....

(a) 67.7 (b) 67.8 (c) 67.75 (d) 68

Third: The range=

(a) 4.4 (b) 4.2 (c) 4.3 (d) 4.5

(4) Which of the following diagrams represents the values 169 , 168 , 178 , 187 ?

(a)

Stem	Leaves
1	6 9
1	6 8
1	7 8
1	8 7

Key | 1 | 69 = 169

(b)

Stem	Leaves
16	8 9
17	8
18	7

Key | 16 | 8 = 168

(c)

Stem	Leaves
16	8 9
1	7 8
1	8 7

Key | 1 | 78 = 178

(d)

Stem	Leaves
16	8 9
17	7
18	8

Key | 1 | 78 = 178

(5)

7. The data given in the stem and leaves diagram are ...

a) 162,1745,1867

b) 162,174,175,186,187

c) 16.2,17.4,17.5,18.6,18.7

d) 1.62,1.74,1.75,1.86,1.87

Stem	Leaves
16	2
17	4 5
18	6 7

Key: 17|4 = 17.4

(6) In the opposite diagram Which of the following statement is in correct?

a) The range of the set $(A) = 30$

b) The median for the set $(B) = 62$

c) the mode of the set $(A) = 43$

d) set A is more variant than set B

Set B	Leaves	Set A
0	4	1 2 3 3
6 3	5	4 5
2 1	6	2 5
7 6 5 2	7	1

Key 3|5|4 means 54 for set a and 53 for set B

(7) The given diagram represents The maximum and minimum temperature in cairo during 5 days , then the difference between the arithmetic mean of the maximum and the arithmetic mean of the minimum = ...

Min: Temperature	Stem	Max: Temperature
6 5	1	8
1	2	2 3
3	3	4
1	4	2

Key | 5 | 1 | 8 means Min : Temperature 15 , and Max : Temperature 18

(a) 2.6 (b) 8 (c) 27.8 (d) 19.8

(8) The data in the given diagram are .....

(a) 162 , 1745 , 1867

(b) 162 , 174 , 175 , 186 , 187

(c) 16.2 , 17.4 , 17.5 , 18.6 , 18.7

(d) 1.62 , 1.74 , 1.75 , 1.86 , 1.87

Stem	Leaves
16	2
17	4 5
18	6 7

Key | 17 | 4 = 17.4

(9)

In the given diagram which of the following statements is false ?

(a) The range of group A = 30

(b) The median of group B = 62

(c) Mode for group (A) is 43

(d) The data in group A is more divergent than in group B

Group B	Stem	Group A
0	4	1 2 3 3
6 3	5	4 5
2 1	6	2 5
7 6 5 2	7	1

Key | 3 | 5 | 4 means 54 for group A and 53 for group B

(10) The given stem and leaf diagram represents the maximum and minimum temperature in ElSharkeya governorate during 5 days, then the difference between the arithmetic mean of the maximum and the arithmetic mean of the minimum equals .....

(a) 2.6 (b) 8 (c) 27.8 (d)19

Maximum	Stem	Minimum
8	1	5 6
3 2	2	1
4	3	3
2	4	1

Key: 8|1|5 means maximum 18 and minimum 15

(1) Represent each of the following data sets using the stem-and-leaf method :

First group	10	26	9	12	27	13	19	15	27	12	29	22
Second group	11	12	10	15	30	9	29	35	11	34	11	12
Third group	1.1	2.4	3	2	6.6	5.8	0.5	2.5	4.1	2.2		

(2)

📖 The opposite data represents the number of mathematics books in the libraries of 15 schools :

It is required to write the original data of the number of books for each school.

Stem	Leaves
0	1 1 1 2
1	0 1 1 1 2 2 3 3 4
2	1 1

Key	1 | 3 = 13

(3) Represent the following data by stem and leaves diagram,

20	8	19	13	5	23	5	17	21	26
14	15	21	12	24	4	21	29	13	7

then show which of the two groups is more diversion?

(4) The following data represents the records of some swimmers have achieved in a 50 meter freestyle swimming race measured in seconds.

25.7 ,27.3, 25.6 , 30 ,31.4 , 28.2 ,29.1
28.6 ,26.4 , 27.3 , 30.6 , 27.4 , 25.9 , 27.6
32, 31.9 , 26.9, 29.1 , 27.3 , 28.4

(1) Represent the data by using stem and leaf
(2) What is the time of the first contestant and the last contestant do need to finish the race

(5) 📖 **The following data represents the heights of 30 students in a secondary school , measured in centimeters :**

161	175	174	165	167	177	180	182
170	157	170	176	185	188	162	165
159	172	158	171	169	173	175	178
164	158	172	170	178	181		

It is required to represent data using the stem and leaf method.

(6) The following data represents the maximum and minimum temperatures for Alexandria city over a period of two weeks.

Maximum temperature	19	28	22	29	25	29	32	35	36	34	37	39	41	42
Minimum temperature	13	22	19	18	16	20	21	22	23	23	21	30	32	31

Represent the temperatures by stem and leaves with description of this date and which of these data is more dispersion

(7) In a study of the time (in minutes) taken to complete a specific task by two groups of workers, their data were as follows:

Group A	32	35	24	36	44	38	40
Group B	43	33	47	37	39	45	41

(1) Draw a double stem-and-leaves plot for this data

(2) Determine which group is faster in completing the task based on the median and range

(8) The following data represents the marks Ahmed and Mahmoud get in a weekly mathematics test. Where the test is out of 50.

Ahmed 37 45 31 36 35 40 41 42 40 45 50 39
Mahmoud 42 23 34 37 41 34 37 45 46 48 45 43

Represents the marks Ahmed and Mahmoud get by using the double stem and leaf. Describe these marks and state which of them more divergent.

(2) Quartiles and their graphic representation

First case : If the number of data n is odd and (n + 1) is divisible by 4

The order of the first (lower) quartile (Q_1)$=\frac{n+1}{4}$

The order of the second (middle) quartile $Q_2 = \frac{n+1}{2}$ (order of the median)

The order of the third (upper) quartile $Q_3 = \frac{3}{4}(n + 1)$

NOTE

First quartile Q_1: It is the value preceded by a quarter of the data (25%) and followed by three quarters of the data

Second quartile Q_2: It is the median, meaning the value that is preceded by $\frac{1}{2}$ of the data (50%) and followed by the other half

Third quartile Q_3: Its the value preceded by $\frac{3}{4}$ the data (75%) and followed by quarter the data (25%)

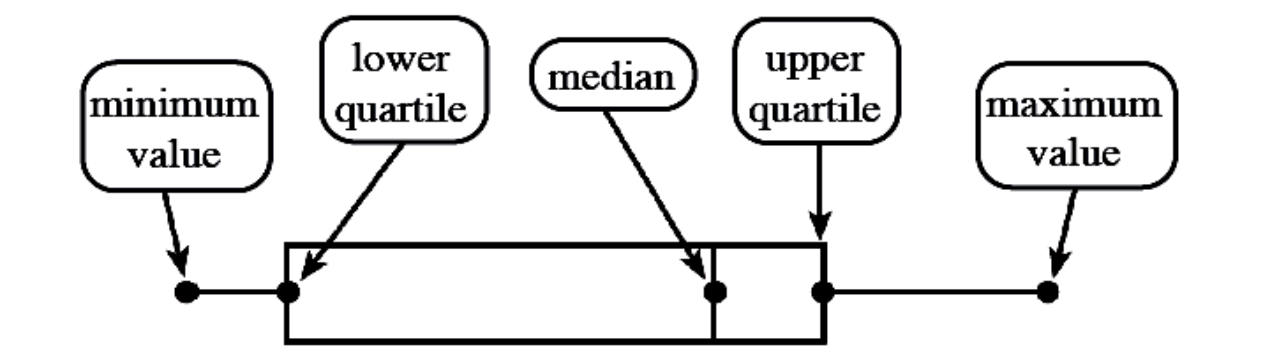
Case 2

If the number of data points is even or odd (n + 1) is not divisible by 4, the quartiles are determined using the following formula:

Required quartile=

its previous value + (its next value - its previous value) (its order - its previous order)

box plot uses these values to describe the data



➢ Semi-interquartile range

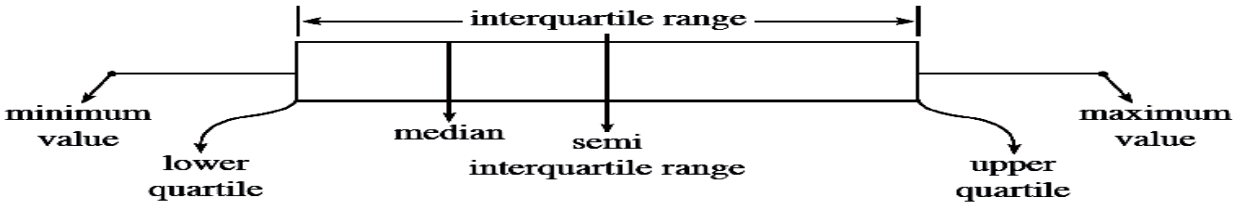
(1)interquartile range$=Q_3-Q_1$

(2)Semi -interquartile range =

$$\frac{upper\ quartile - lower\ quartile}{2} = \frac{Q_3-Q_1}{2}$$

where : Q " semi interquartile range "

Q_1 lower quartile Q_3 upper quartile



Example(1) : Find the three quartiles of the following set of data : 23 , 7 , 16 , 2 , 4 , 5 , 1 , 21 , 15 , 22 , 14 , 11 , 10 , 9 , 3

Solution:

First step :

Arrange : 1 , 2 , 3 , 4 , 5 , 7, 9 ,10, 11 , 14 , 15 , 16 , 21 , 22 , 23

Second step:

numbers of data n = 15 and n + 1 = 16 (divisible by 4)

the order of first quartile $Q_1 = \frac{15+1}{4} = 4$, its value = 4

The order of the second quartile $Q_2 = \frac{15+1}{2} = 8$,Its value = 10

The order of the third quartile $Q3 = \frac{3}{4}(n + 1) = 12$

,its value of $Q_3 = 16$

Example(2) : The interquartile range of the values 3 , 5 , 2 , 9 , 3 , 2 , 5 , 6 ,7 , 8 , 4 equals .....

(a) 2 (b) 4 (c) 5 (d) 7

Solution

Arrange : 2 , 2 , 3 , 3 , 4 , 5 , 5 ,6 , 7 , 8 , 9

numbers of data $n = 11$ and $n + 1 = 12$ (divisible by 4)

the order of first quartile $Q_1 = \frac{11+1}{4} = 3$, its value = 3

The order of the third quartile $Q3 = \frac{3}{4}(n + 1) = 9$

,its value of $Q_3 = 7$

interquartile range=$Q_3 - Q_1$=7-3=4

Example(3) :try to solve

(1)find the lower , middle , upper quartiles of the following set of values : 70, 81, 82, 58, 88, 90, 93

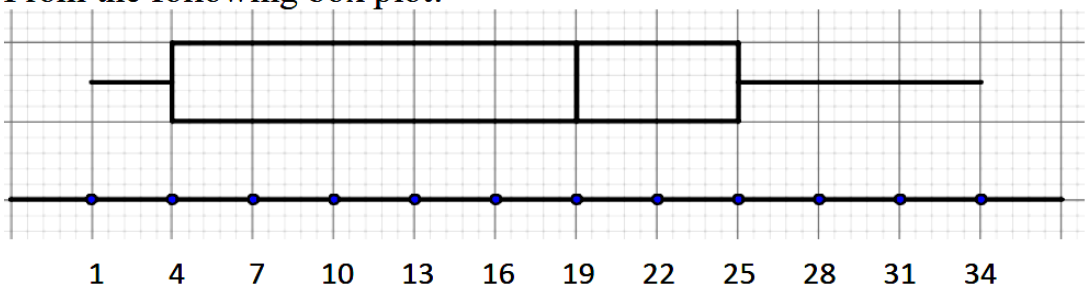
(2) Find the three quartiles of the following values 4,17 ,9 , 22 , 15 ,8 ,6, 11 , 18 , 19 , 22

then represent this data by using the box plot and find the semi-interquartile range

Solution

Example(4) :

From the following box plot:



the median =……..

(a) 4 (b) 14.5 (c) 19 (d) 25

Solution

Median =19

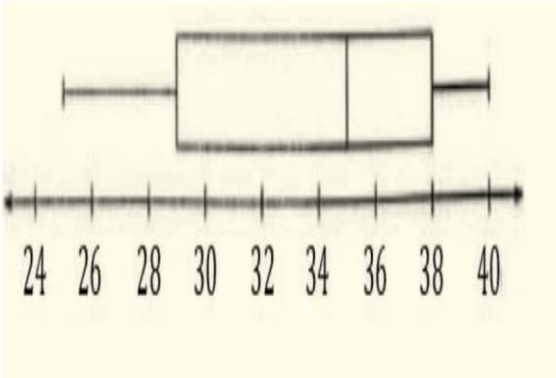
Example(5) :

in the given box plot

The semi quartile range=

(a) 15 (b) 7.5

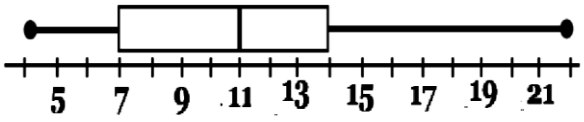
(c) 9 (d) 4.5



Solution

Semi -interquartile range $= \frac{Q_3 - Q_1}{2} = \frac{38-29}{2} = 4.5$

Example(6) :



in the given box plot

The interquartile range=.....

(a) 7 (b) 14 (c) 3.5 (d) 18

Solution try to solve

interquartile range=$Q_3 - Q_1$=14-7=7

Example(7) The value that is followed by 75 % of data approximately is called

(a) lower quartile. (c) upper quartile

(b) median.. (d) interquartile range

Solution

(a) lower quartile

Example(8) The value that is preceded by $\frac{3}{4}$ of data approximately is called

(a) lower quartile. (b)median.

(c) upper quartile. (d) interquartile range.

Solution

(c) upper quartile.

Example(9) If the order of Q_1, is 5.75 then the number the values is

(a) 23 (b) 22 (c) 24 (d) 21

Solution

$$\frac{n+1}{4} = 5.75 \quad then \quad n = 22$$

Examples on case (2)

Example(1) : The data represented by the stem -and- leaves method represents the ages of 10 persons who visit a library on a day. Find the three quartiles of these data.

stem	leaves
1	2 3
2	0 2 2 4 5 5
3	1 3
	Key 24 = 2|4

Solution:

The ascending order of the values :

12 , 13 , 20 , 22 , 22 , 24 , 25 , 25 , 31 , 33

Order of the first quartile (lower) = $\frac{n+1}{4}$ = 2.75 The value of the first quartile (Q_1) =

13 + (20 – 13) (2.75 – 2) = 13 + 5.25 = 18.25

Order of the 2nd quartile (middle) = $\frac{n+1}{2} = 5.5$

The value of the second quartile (Q_2)

= 22 + (24 – 22) (5.5 – 5) = 23

Order of the 3rd quartile (upper) $= \frac{3}{4}(n + 1) = 8.25$

The value of the third quartile (Q_3) = 25 + (31 – 25) (8.25 – 8) = 26.5

Example (2) : try to solve

Find the three quartiles

Lower-middle -upper)

Of the opposite figure

stem	leaves
0	6 7 5
1	9 0 1 3 8 2 2 1 2
2	5 1 0 1 key19=1/9

Example (3)

The third quartile of the values : 1 ,4,3, 7,8,5 ,9 ,2 is

(a) 3.75 (b) 3 (c) 7.75 (d) 5.5

Solution

Arrange : 1 , 2 , 3 , 4 ,5 ,7 , 8 ,9

Order of the 3rd quartile (upper) $= \frac{3}{4}(n + 1) = 6.75$

The value of the third quartile (Q_3) = 7 + (8 – 7) (6.75 – 6) = 7.75

Example (3)

The following double stem and leaf diagram shows the number of working hours of workers in two The following double stem and leaf diagram shows the number of working hours of workers in two factories calculate the three quartiles and the semi-interquartile range of working hours , then draw the box plot for each factory

Second Factory	stem	First Factory
8 6 4 4 2 2	3	5 6 6 6 7
9 6 3 1 1	4	0 1 3 5 8 9
5 4 2 2	5	1 1 2 2 4 6
2 1 1	6	1 2

Solution try to solve

H.W(2)

(1)The median of the values:3 ,2 ,1 ,4 ,6 is
(a) 1 (b) 2 (c) 3 (d) 4

(2) If the students' test scores are as follows:
95,90,85,80,75,70,65,60,55 The median for this data is .....
(a) 70 (b) 75 (c) 80 (d) 85

(3)The lower quartile of the values :7 ,4 ,3,11 ,9, 8 ,2 is
(a) 7 (b) 3 (c) 9 (d) 3.5

(4)The first quartile of the values :26 ,22 ,28,32 ,24, 30,20 is
(a) 22 (b) 24 (c) 26 (d) 30

(5) The third quartile of the values : 1 ,4,3, 7,8,5 ,9 ,2 is
(a) 3.75 (b) 3 (c) 7.75 (d) 5.5

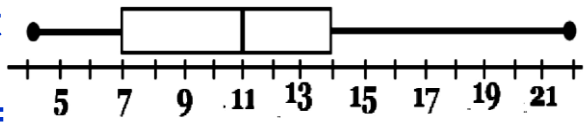
(6) The upper quartile of the values :
14,24,16, 12,18,20 ,24 ,16,26,13,27 is .....
(a) 24 (b) 20 (c) 26 (d) 18

(7) If the order of the upper quartile of a set of single values is 48 -then the number of these individuals is .....
(a) 64 (b) 60 (c) 96 (d) 63

(8) If the order of Q_1, is 5.75 then the number the values is
(a) 23 (b) 22 (c) 21 (d) 24

(9) If the order of Q_1, is 3.75 and it lies between 37 and 41 then $Q_1 =$
(a) 39.5 (b) 40.5 (c) 38.5 (d) 40

(10) in the given box plot



The interquartile range=
(a) 7 (b) 14 (c) 3.5 (d) 18

(11) If $Q_1 = 5$, $Q_2 = 7$, $Q_3 = 12$,
then the semi-interquartile range =
(a) 7 (b) 1 (c) 2.5 (d) 3.5

(12) If the semi-interquartile range of a set of values is 9 ,
then $Q_3 - Q_1 = \cdots$
(a) 5 (b) 9 (c) 18 (d) 4.5

(13)The interquartile range of the values
3 , 5 , 2 , 9 , 3 , 2 , 5 , 6 ,7 , 8 , 4 equals .....
(a) 2 (b) 4 (c) 5 (d) 7

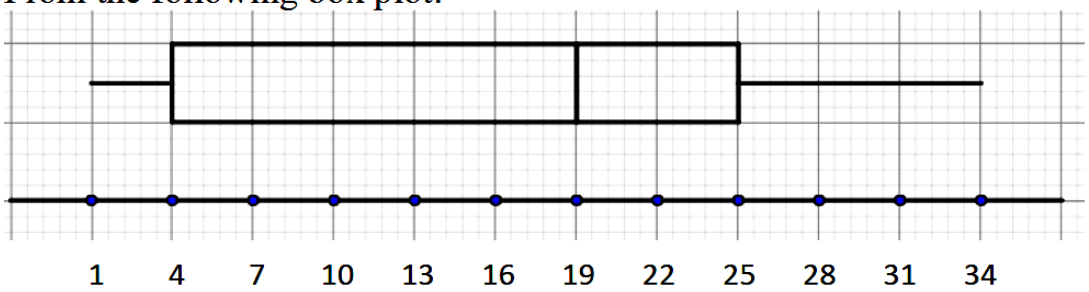
(14) From the following data

stem	leaf					
5	1	4	6	7	8	
6	5	5	5	6	9	9
7	0	1	2	3	4	4

Key: 6|5 means 65

The lower quartile equals .....
(a) 57 (b) 57.5 (c) 58 (d) 66

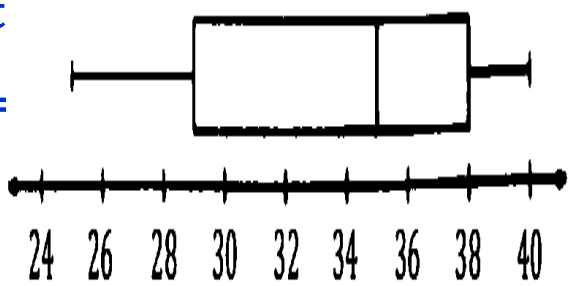
(15)
From the following box plot:



the median =.....
(a) 4 (b) 14.5 (c) 19 (d) 25

(16)in the given box plot

The semi quartile range=

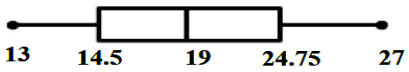


(a) 15 (b) 7.5

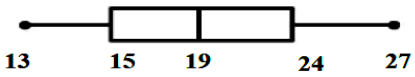
(c) 9 (d) 4.5

(16) The box plot of the following:27,24,20,18,15,13is

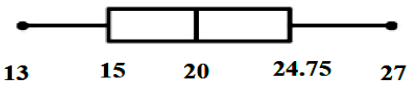
(a)



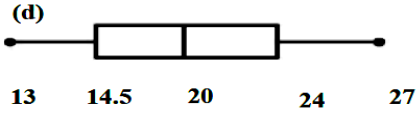
(b)



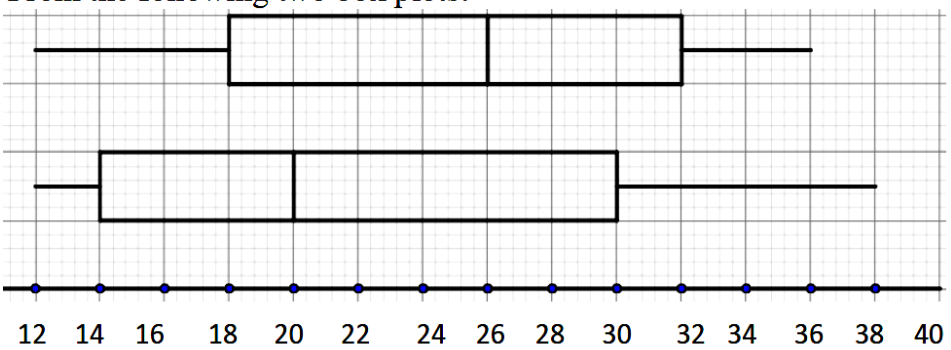
(c)



(d)



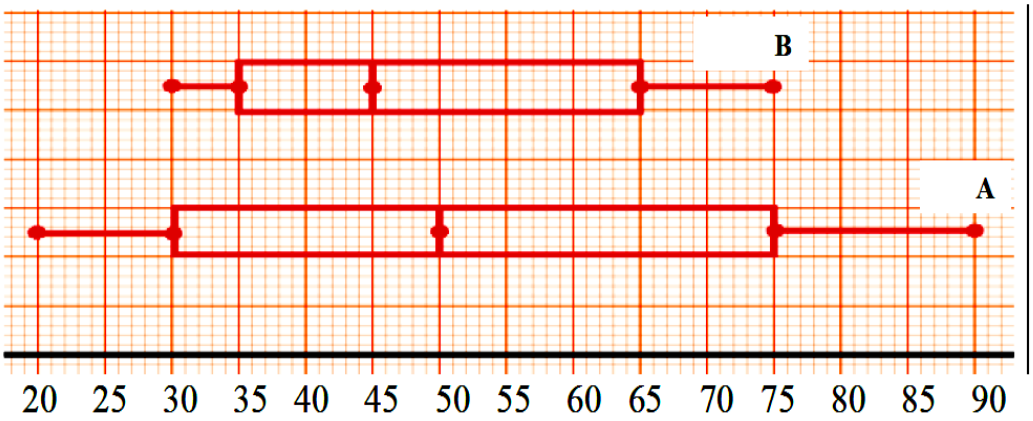
(17) From the following two box plots:



The difference between the interquartile range of the two groups equals

(a) 2 (b) 4 (c) 6 (d) 8

(18)The following figure shows the distribution of marks for two exams for a group of students , then .....



(a) The median mark of the exam(A) less than the median mark of exam(B).
(b) The interquartile range of the marks of exam (B) greater than the interquartile range of the marks of exam (A).
(c) The lower quartile of the marks of exam (B) equal the lower quartile of the marks of exam (A).
(d) The marks of exam (A) more divergent than the marks of exam (B).

(19)The value that is followed by 75 % of data approximately is called

(a) lower quartile. (c) upper quartile

(b) median.. (d) interquartile range

(20)The value that is preceded by $\frac{3}{4}$of data approximately is called

(a) lower quartile. (b)median.

(c) upper quartile. (d) interquartile range.

(21)If the order of Q_1, is 5.75 then the number the values is
(a) 23 (b) 22 (c) 24 (d) 21

(1)find the lower , middle , upper quartiles of the following set of values : 70, 81, 82, 58, 88, 90, 93

(2) Find the three quartiles of the following values
4,17 ,9 , 22 , 15 ,8 ,6, 11 , 18 , 19 , 22
then represent this data by using the box plot and find the semi-interquartile range

(3)The following is the amount of daily milk production in liters for a sample of cows selected from a farm:
30, 27, 18, 20, 29, 34, 25, 32, 29, 21, 23, 28, 25, 19
Represent the data using the stem and leaf method and calculate the semi-interquartile range

(4) Draw the box blot of the following data :

a) 27, 24, 20, 18, 17, 15, 13

b)

Stem	Leaves
4	0 3 3 6 7
5	1 8 9
6	2 3 4

Key ← 51 = 511

(5) The following table shows the marks of 20 students in mathematics:

92	78	73	89	86	85	76	81	73	88
83	75	83	83	71	86	82	94	100	98

Represent this data by stem-and-leaves diagram and calculate the semi-interquartile range.

(6) The following marks represent the marks of 15 students in the statistics exam

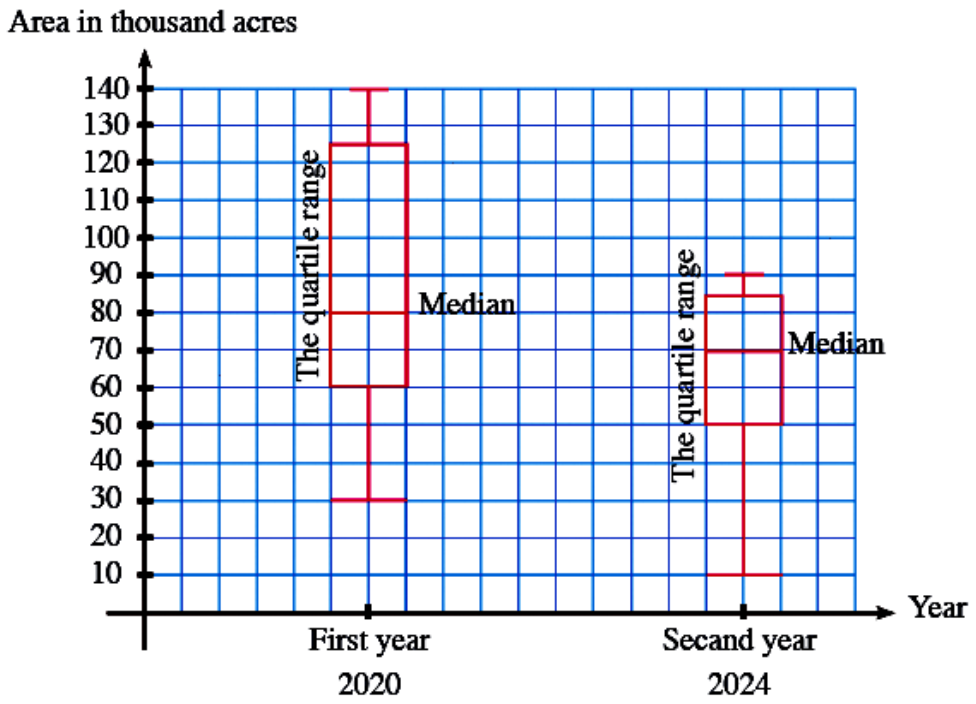
37	40	45	23	18
44	53	38	49	55
15	58	35	32	42

Draw the box plot these marks

(7)The following graph represents the cultivated area in thousand acres in 25 villages during two different years.

(a) Find the upper quartile, the lowest quartile, and the median. And the semi- interquartile range of the two years

(b) What do you conclude from this data?



Finding quartiles from frequency tables:

The first step: We create the ascending cumulative table
The second step: We determine the orders of the quartiles
First quartile order $= \frac{n}{4}$
second quartile order $= \frac{2n}{4}$
third quartile order $= \frac{3n}{4}$

Third step : determine the interval (class) which the required quartile lies in
(it called the quartile interval) from this interval we determine the start of the interval , its width Interval frequency, ascending cumulative frequency at previous of interval

Forth step : Use the flowing formula to calculate the required quartile
The value of the required quartile =

$$\text{Quartile interval start} + \frac{\text{Quartile order} - \text{cum.freqat start ofthe interval}}{\text{frequency corresponding to the quartile interval}} \times \text{interval width}$$

Example(1) :

The following frequency table shows the number of working hours in a week for 50 workers in a factory ,then find the three quartiles

Number of working hours	22-	27-	32-	37-	42-	47-	total
frequency	9	3	10	8	12	8	50

Solution

Frequency ascending cumulative table	
upper boundaries	Ascending cumulative frequency
Less than 22	Zero
Less than 27	9
Less than 32	12
Less than 37	22
Less than 42	30
Less than 47	42
Less than 52	50

(1) Assigning the first quartile Q_1:

order of $Q_1 = \frac{50}{4} = 12.5$

$\therefore$ Q_1 lies in the interval 12 , 22

$$Q_1 = 32 + \frac{12.5 - 12}{10} \times 5 = 32.25$$

(2) Assigning the second quartile (median) Q_2

order of $Q_2 = \frac{50}{2} = 25$

$$\therefore\ Q_2 = 37 + \frac{25 - 22}{8} \times 5 = 38.875$$

(3) Assigning the third quartile Q_3

Order of $Q_3 = \frac{50 \times 3}{4} = 150$

$$Q_3 = 42 + \frac{37.5 - 30}{12} \times 5 = 45.125$$

Example(2) : The following table shows the numbers of days off that 50 employee have in governmental organization last year.

Number of days off	0-	10-	20-	30-	40-	total
frequency	6	8	15	12	9	50

Find the three quartiles and the semi -interquartile range.

Solution

Frequency ascending cumulative table	
upper boundaries	Ascending cumulative frequency
Less than 0	Zero
Less than 10	6
Less than 20	14
Less than 30	29
Less than 40	41
Less than 50	50

(1) Assigning the first quartile Q_1:

order of $Q_1 = \frac{50}{4} = 12.5$

$\therefore$ Q_1 lies in the interval 6 , 14

$$Q_1 = 10 + \frac{12.5 - 6}{8} \times 10 = 18\frac{1}{8}$$

(2) Assigning the second quartile (median) Q_2

order of $Q_2 = \frac{50}{2} = 25$, $\therefore$ Q_2 lies in the interval 14 , 25

$$\therefore\ Q_2 = 20 + \frac{25 - 14}{15} \times 10 = 27\frac{1}{3}$$

(3) Assigning the third quartile Q_3

Order of $Q_3 = \frac{50 \times 3}{4} = 37.5$,Q_3 lies in the interval 29,41

$$Q_3 = 30 + \frac{37.5 - 29}{12} \times 10 = 37\frac{1}{12}$$

Example 3

If the following table represent the frequency table of the average of the hemoglobin level in blood in a sample of 50 persons find the quartiles algebraically and graphically

Hemoglobin level	13-	14-	15-	16-	17-	18-	total
frequency	3	5	15	16	10	1	50

Solution try to solve

Second : Finding quartiles graphically:

If the frequency distribution of temperatures during 60 consecutive days in the spring season in Egypt, as follows:

Temperatures	16-	18-	20-	22-	24-	26-	28-	Total
Number of days (frequency)	**4**	**7**	**10**	**18**	**9**	**7**	**5**	60

Find the quartiles graphically

Solution

n = 60

The order of the first quartile Q_1

$= \frac{60}{4} = 15$

The order of the second quartile

$Q_2 = \frac{60}{2} = 30$

The order of the third quartile $= \frac{3\times60}{4} = 45$

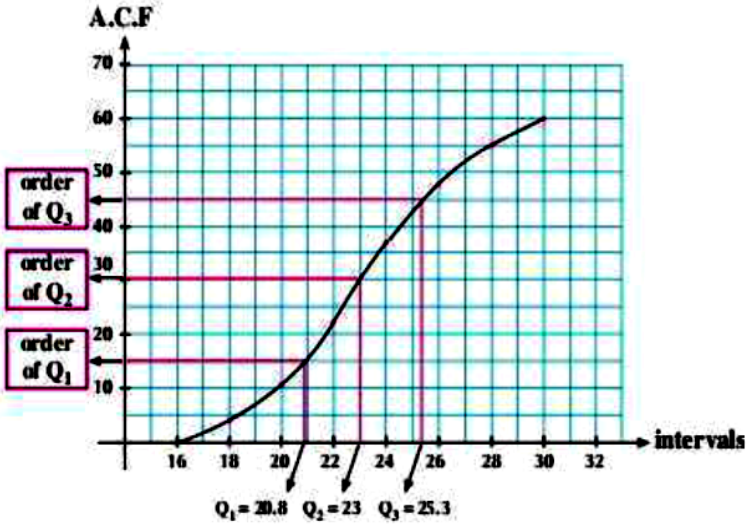
frequency ascending cumulative table	
upper boundaries	ascending cumulative frequency
Less than 16	Zero
Less than 18	4
Less than 20	11
Less than 22	21
Less than 24	39
Less than 26	48
Less than 28	55
Less than 30	60

From the opposite graph

$Q_1 = 20.8$

$Q_2 = 23$

$Q_3 = 25.3$



H.W(3)

(1) From the opposite frequency asending cumulative table, choose the correct answer from the given ones :

Interval	Frequency	Upper boundaries	The cumulative frequency
4 –	2	less than 4	0
8 –	4	less than 8	2
12 –	8	less than 12	6
16 –	6	less than 16	14
20 –	4	less than 20	20
Total	24	less than 24	24

(1) The order of Q_1 = ...

(a) 6.45 (b) 7

(c) 6 (d) 4

(2) The quartile interval of the lower quartile is .....

(a) 4 – (b) 8 –

(c) 12 – (d) 16 –

(3) The value of Q_1 = ...

(a) 14 (b) 12

(c) 13 (d) 15

(4) The order of Q_2 = ...

(a) 12.5 (b) 12 (c) 15 (d) 16

(5) The value of Q_2 = ...

(a) 12 (b) 13 (c) 14 (d) 15

(6) The order of Q_3 = ...

(a) 15 (b) 17 (c) 18 (d) 20

(7) The value of Q_3 = ...

(a) $18\frac{1}{2}$ (b) $18\frac{3}{4}$ (c) $18\frac{2}{3}$ (d) 18

(8) The semi-interquartile range = ...

(a) 4.5 (b) $3\frac{1}{3}$ (c) 3.5 (d) $4\frac{2}{3}$

(2) The following table shows the frequency table of the ages of 20 teachers:

The ages	33-	38-	43-	48-	53-	total
Number of teachers	3	7	4	2	4	20

Then the semi-interquartile range of these ages =.....

) $50\frac{1}{2}$ (b) $39\frac{3}{7}$ (c) $5\frac{15}{28}$ (d) 43

(3)

The semi-interquartile range =.....

(a) 4.5 (b) 9

(c) 26 (d) 35

upper limit	ascending cumulative frequency
less than 15	0
less than 20	3
less than 25	12
less than 30	27
less than 35	45
less than 40	57
less than 45	60

(4) The following data shows a frequency table for the ages of 20 teachers

Ages	33-	38-	43-	48-	53-	Total
Number of teachers	3	7	4	2	4	20

Calculate the semi interquartile range for these ages

(5) If the following table represent the frequency table of the average of the hemoglobin level in blood in a sample of 50 persons find the quartiles algebraically and graphically

Hemoglobin level	13-	14-	15-	16-	17-	18-	total
frequency	3	5	15	16	10	1	50

(6) The following table shows the numbers of days off that 50 employee have in governmental organization last year.

Number of days off	0-	10-	20-	30-	40-	total
frequency	6	8	15	12	9	50

Find the three quartiles and the semi -interquartile range.

(7) The following frequency table shows the weights of a number of births during 14 days in a hospital

Birth weights in kilograms	2–	2.5–	3–	3.5–	4–	4.5–	Total
Number of births	3	7	10	8	4	2	34

Find the inter-quartile range and the semi- interquartile range

(8) The following table shows the heights of 240 female students at a university

Length in cm	140–	145–	150–	155–	160–	165–	170–	175–	180–	Total
Number of students	3	10	21	54	72	48	25	5	2	240

Find semi- interquartile range and represent the data by a box plot

(9) Energy : In a study of the consumption of a group of gasoline-powered cars, the results were as follows :

Number of kilometers per each liter	20-	25-	30-	35-	40-	45-
Number of cars	7	11	12	7	6	8

form the ascending cumulative frequency table and then find the quartiles in two different ways.